Packet Switching Delay – Student Worksheet

Course: Data Communications & Computer Networks

Problem Statement

A source node **S** wants to send a message of size **6000 bytes** to a destination node **D** using **packet switching**. The message is divided into **3 equal-sized packets**. The packets travel through two intermediate routers (**R1** and **R2**) before reaching the destination.

Network Parameters

```
- Link data rate: 2 Mbps (all links)
- Propagation delay per link: 10 ms
- Processing delay at each router: 2 ms
- Queueing delay: Negligible
- Packet headers: Ignored
- Switching technique: Store-and-forward packet switching
```

Network Diagram

```
        Link 1        Link 2        Link 3
     2 Mbps, 10 ms  2 Mbps, 10 ms  2 Mbps, 10 ms

   +----+        +----+        +----+        +----+
   | S  | -----> | R1 | -----> | R2 | -----> | D  |
   +----+        +----+        +----+        +----+
                    ^              ^
              Proc = 2 ms     Proc = 2 ms
```

Student Tasks

```
1. Determine the size of each packet in bytes and bits.
2. Calculate the transmission delay for one packet on one link.
3. Calculate the end-to-end delay for the first packet.
4. Calculate the total time required for all packets.
5. Briefly explain why packet switching is efficient in this scenario.
```

Instructor Solution

Step 1: Packet Size

Total message size = 6000 bytes, Number of packets = 3. Each packet size = 2000 bytes = 16,000 bits.

Step 2: Transmission Delay

Transmission Delay = Packet size / Link rate = 16,000 / (2 × 10■) = 0.008 s = 8 ms.

Step 3: End-to-End Delay (First Packet)

Each link contributes 8 ms transmission + 10 ms propagation. There are 3 links and 2 routers with 2 ms processing delay each. Total delay = 3 × (8 + 10) + 2 × 2 = 58 ms.

Step 4: Total Time for All Packets

Due to pipelining, additional packets add only one transmission delay each. Total time = 58 + (3 – 1) × 8 = 74 ms.

Conceptual Explanation

Packet switching allows multiple packets to be in transit simultaneously on different links. This pipelining improves link utilization and reduces overall transmission time compared to circuit switching.